

# Product-Probe Drift, Kronecker Nyström Approximation, and Trace Estimation

Residual progress from moments near order one

29 September 2026

## Abstract

A product probe can make useful progress without embedding a prescribed subspace. We prove that one fresh spherical or Gaussian tensor probe reduces the trace of every positive-semidefinite residual  $R$  by at least  $\text{tr}(R^2)/(\kappa \text{tr } R)$  in expectation. The constant  $\kappa$  is the exponential of a directional moment slope at order one, improving the usual fourth-moment constant in every tensor factor. This gives Nyström error bounds for arbitrary eigenvectors, including a relative trace-tail guarantee and Schatten-norm decay. Combined with Holden’s diagonal-control estimator, it yields a nonadaptive real product-query bound of order  $(N\kappa/\varepsilon^2)^{1/3}$ , with  $q + 1$  calls per complex probe. A spike-plus-flat example brackets the optimal drift exponent between a Chernoff rate and the order-one slope. We also give independent-factor, fresh-row, and CountSketch-block versions, while leaving the unrestricted minimax trace-query complexity open.

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# 1 Introduction and main results

## 1.1 The query model

A query to an unknown real symmetric matrix  $A \succeq 0$  must be a rank-one tensor:

$$(x_1, \dots, x_q) \mapsto A(x_1 \otimes \dots \otimes x_q), \quad x_j \in \mathbb{R}^{n_j}, \quad N = \prod_j n_j. \quad (1.1)$$

The oracle returns the entire vector. We count oracle calls; exact arithmetic and storage are not charged. For trace estimation the goal is to output  $\hat{T}$  with  $|\hat{T} - \text{tr } A| \leq \varepsilon \text{tr } A$  with probability at least  $2/3$ , for every fixed input, and we then take  $n_j = n$ ,  $N = n^q$ ,  $0 < \varepsilon < 1/2$ . Meyer, Swartworth and Woodruff formulated Kronecker matrix–vector complexity and list tight Kronecker trace estimation as open [1]; the catalogue problem RA-11 asks for the joint dependence on  $(n, q, \varepsilon)$  [3]. Meyer and Avron showed that Hutchinson-type averaging is exponentially bad in  $q$  [2].

A full response  $Az$  can do more than a scalar  $z^*Az$ : it can build a PSD Nyström approximation  $\hat{A}$  whose trace is known exactly, after which only the residual trace must be estimated. The benefit depends on how much residual mass each legal probe removes. This *progress*, rather than an embedding of a prescribed subspace, is the central object of the paper.

## 1.2 Probe laws and the order-one constant

A *real probe* is  $z = x_1 \otimes \cdots \otimes x_q$  with  $x_j$  independent and uniform on the real sphere of radius  $\sqrt{n_j}$ . A *complex probe* uses the complex spheres of the same radii; for a real matrix it supplies the real two-column block  $W = [\Re z \ \Im z]$ , whose columns are dependent and are always analyzed together. For a real probe put  $W = z$ . Gaussian factors give exactly the same Nyström outputs (Section 2), so real probes are the columns of a Gaussian Khatri–Rao sketch. Define

$$D_n^{\mathbb{R}} = \log n + \psi(3/2) - \psi(n/2 + 1), \quad \kappa_{\mathbb{R}} = \exp\left(\sum_{j=1}^q D_{n_j}^{\mathbb{R}}\right), \quad (1.2)$$

$$D_n^{\mathbb{C}} = \log n + 1 - H_n, \quad \kappa_{\mathbb{C}} = \exp\left(\sum_{j=1}^q D_{n_j}^{\mathbb{C}}\right), \quad (1.3)$$

with  $\psi = (\log \Gamma)'$  and  $H_n = \sum_{a \leq n} a^{-1}$ . Write  $\kappa$  for the constant of the chosen law. For  $n = 2$ ,  $e^{D_2^{\mathbb{R}}} = e/2 = 1.35914$  and  $e^{D_2^{\mathbb{C}}} = 2/\sqrt{e} = 1.21306$ , against fourth-moment constants  $3/2$  and  $4/3$ .

For a sketch  $Z$  put  $\mathcal{N}_A(Z) = AZ(Z^{\top}AZ)^{\dagger}Z^{\top}A$ , let  $\hat{A}_m$  be the Nyström approximation built from  $m$  independent probes (or blocks),  $R_m = A - \hat{A}_m$ ,  $T = \text{tr } A$  and  $\tau_r = \tau_r(A) = \sum_{i > r} \lambda_i(A)$ .

## 1.3 Main results

**Theorem A (order-one drift)** (Theorem 4.2). *For every fixed nonzero real  $R \succeq 0$  and one fresh probe or block  $W$ , with  $R' = R - RW(W^{\top}RW)^{\dagger}W^{\top}R$ ,*

$$0 \preceq R' \preceq R, \quad \mathbb{E}(\text{tr } R - \text{tr } R') \geq \frac{\text{tr}(R^2)}{\kappa \text{tr } R}.$$

*More generally, for independent isotropic factor laws with directional moment functions  $m_j(p)$ , the same holds with  $\kappa = \lim_{\beta \downarrow 0} \prod_j m_j(1 + \beta)^{1/\beta}$ .*

**Theorem B (unconditional Kronecker Nyström)** (Theorems 5.2, 5.4, 5.5). *For every real PSD input  $A$ , with no hypothesis on its eigenvectors:*

(i) *The Frobenius residual satisfies*

$$\mathbb{E}\|A - \hat{A}_m\|_F^2 \leq \frac{\kappa T^2}{m + \kappa T^2/\|A\|_F^2} \leq \frac{\kappa T^2}{m}.$$

(ii) *Let  $1 \leq r < \text{rank } A$  and  $0 < \varepsilon \leq 1$ . The budget*

$$m \geq \left\lceil \kappa r \left[ \varepsilon^{-1} + \log(\varepsilon^{-1}) + \log(T/\tau_r) \right] \right\rceil$$

*ensures  $\mathbb{E} \text{tr}(A - \hat{A}_m) \leq (1 + \varepsilon)\tau_r$ . This concerns the full, untruncated approximation.*

(iii) *For every  $p > 1$ ,*

$$\mathbb{E} \text{tr } R_m^p \leq T^p \left( 1 + \frac{m}{\kappa(p-1)} \right)^{-(p-1)}.$$

(iv) *For  $0 < \mu \leq T$ ,*

$$\mathbb{P}\{\|R_m\| \geq \mu\} \leq \min \left\{ 1, \frac{2T}{\mu} e^{-m\mu/(2\kappa T)} \right\}.$$

*In particular,  $m \geq (2\kappa T/\rho) \log(2T/(\rho\delta))$  makes  $\hat{A}_m + \rho I$  a factor-two preconditioner for  $A + \rho I$ , with probability at least  $1 - \delta$ , when  $0 < \rho \leq T$ .*

**Theorem C (trace estimation with real product queries)** (Theorem 6.4). *Choose one of the two pairs*

$$(b, \kappa) = (1, e^{qD_n^{\mathbb{R}}}) \quad \text{or} \quad (b, \kappa) = (q+1, e^{qD_n^{\mathbb{C}}}).$$

*There is a nonadaptive unbiased estimator using  $bm + 2s$  real product queries with*

$$\text{Var } \hat{T} \leq \frac{2N\kappa T^2}{ms^2}.$$

*Set  $L_b = (6bN\kappa/\varepsilon^2)^{1/3}$ . At success probability  $2/3$ , the query budget is*

$$\mathcal{D}_b = \min\{N, b\lceil L_b/b \rceil + 2\lceil L_b \rceil\} \leq \min\{N, 3L_b + b + 2\}.$$

*We take the smaller of the real and complex budgets. For  $n = 2$ , the real branch has the smaller leading term exactly when  $q \leq 30$ . In the complex implementation, retaining the  $q+1$  interpolation vectors is pathwise at least as accurate as keeping only the two-column block.*

**Theorem D (domination of benchmark budgets)** (Theorem 7.1). *The rounded drift budget is at most either benchmark budget defined in Section 7, for every admissible  $(n, q, \varepsilon)$ . Before the exact basis cap, their leading terms exceed the drift leading term by factors of at least 18.44 and 44.23, respectively. For complex binary factors, the optimized fractional benchmark also has an  $\exp(\Theta(\sqrt{q}))$  coefficient overhead (Proposition 7.2); the drift bound has a polynomial prefactor at  $\varepsilon = (9/10)^q$ .*

**Theorem E (the sharp exponent is bracketed)** (Theorem 8.2). *Let  $\kappa_{\text{opt}}(n, q)$  be the best constant in the drift inequality of Theorem A for the real one-column law or the complex block law, and  $\gamma_n = -\min_{0 \leq \lambda \leq 1} \log c_n(\lambda)$ , where  $c_n$  is the local moment function of the same field. If  $n^q \geq 4$  then*

$$\frac{1}{32} e^{q\gamma_n} \leq \kappa_{\text{opt}}(n, q) \leq e^{qD_n}.$$

*For  $n = 2$  complex,  $\gamma_2 = \log 2 - 1 - \log \log 2 = 0.059660$  and  $D_2^{\mathbb{C}} = 0.193147$ ; for  $n = 2$  real,  $[\gamma_2^{\mathbb{R}}, D_2^{\mathbb{R}}] = [0.109117, 0.306853]$ .*

If the lower end is the truth (Conjecture 8.4), the binary running base drops from 1.441482 to 1.378748; the benchmark obtained by this analysis with a subexponential drift constant is  $(200/81)^{1/3} = 1.351600$ .

**Theorem F (other probe laws)** (Section 9). *The progress method also gives the following sufficient constants:*

- (i) *Independent isotropic sub-Gaussian factors satisfying  $\mathbb{E}e^{\langle x_j, u \rangle^2 / K_j^2} \leq 2$  have  $\kappa \leq \prod_j 2K_j^2(K_j^2 - 1)$ .*
- (ii) *Independent fresh SRHT rows have the dimension-uniform bound  $\kappa_0 = e^{2-\gamma}/2 = 2.074328$ , where  $\gamma$  is Euler's constant.*
- (iii) *A fresh  $b$ -row CountSketch block has  $\kappa_b = 1 + 2/b$ , counting the whole block as one update.*

*The fresh-row guarantee requires independent sign redraws. The block bound can lose a factor of order  $b$  on a flat residual.*

## 1.4 Roadmap

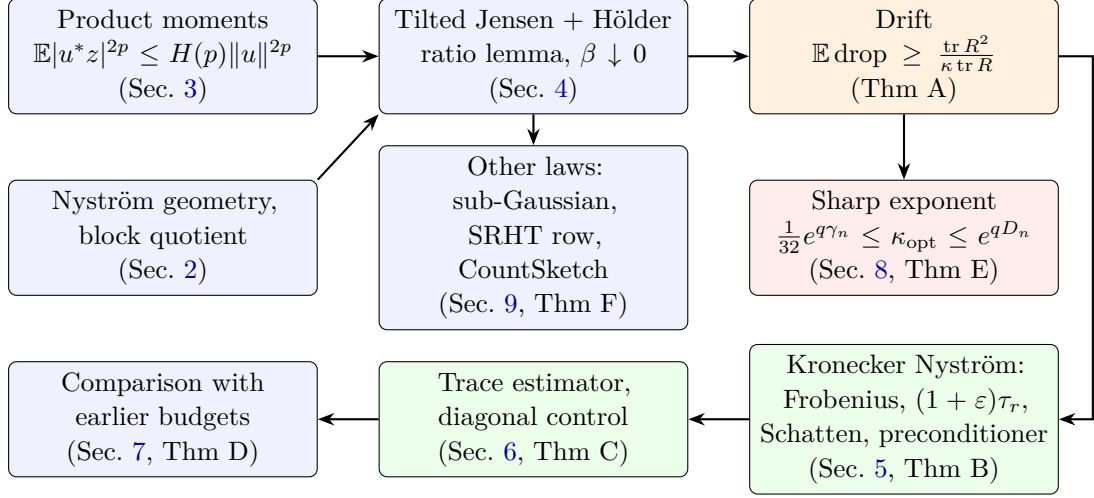


Figure 1: Logical structure of the paper.

## 1.5 Prior work and the contribution

The full-vector Kronecker oracle and its distinction from scalar quadratic-form access are due to Meyer–Swartworth–Woodruff [1]. Meyer–Avron [2] analyze product-probe Hutchinson estimators, including the advantage of spherical and complex factors. Their estimator-specific lower bounds do not determine the complexity of the full-vector model.

Holden’s public manuscript [4] already gives the complex order-one constant  $D_n^{\mathbb{C}}$ , fractional empirical-average bounds, real interpolation of complex responses, diagonal control, and finite angular laws. We use these constructions with explicit attribution. Sections 2–6 reproduce the arguments needed here, so the present guarantees do not depend on a private note.

The new step proposed here is Lemma 4.1: a numerator-weighted change of measure converts moments arbitrarily close to order one into a uniform expected quotient bound. It yields progress for every PSD residual, rather than an embedding guarantee for a fixed subspace. The resulting Nyström analysis uses the standard projection identity and the residual-contraction mechanism of Chen–Epperly–Tropp–Webber [5]. The convex-potential and ramp arguments adapt Epperly [6, Lemma 4.2]; they are not claimed as new matrix inequalities. Divan–Gilles [7] give recent convergence results for randomized and approximate pivoting in a different sampling model.

Camaño–Epperly–Meyer–Tropp [8] obtain structured Nyström guarantees through oblivious subspace injections. Saibaba–Verma–Ballard [9] and Beretta–Musco [10] develop Khatri–Rao low-rank and subspace-embedding analyses. Our contribution is the explicit order-one progress bound, its application to untruncated approximation and trace estimation, and the spike-plus-flat exponent bracket. These are comparisons of proof mechanisms and guarantees; no sharp OSE result or complete resolution of RA-11 is asserted. The literature search is current to 29 September 2026 and does not certify priority.

**Notation and scope.** We use  $\|\cdot\|$  for the Euclidean norm on vectors and the operator norm on matrices;  $\|\cdot\|_F$  is the Frobenius norm. Eigenvalues of a PSD matrix are ordered decreasingly, and  $M^\dagger$  denotes the Moore–Penrose pseudoinverse. Expectations are over the probes, with the

input matrix fixed. All oracle and interpolation statements use exact arithmetic. The bounds count calls, not storage or arithmetic work.

## 2 Nyström geometry and sequential updates

If  $P_Z$  is the orthogonal projector onto  $\text{range}(A^{1/2}Z)$ , then

$$\mathcal{N}_A(Z) = A^{1/2}P_ZA^{1/2}, \quad (2.1)$$

since  $B(B^\top B)^\dagger B^\top$  is the range projector of  $B = A^{1/2}Z$ . Thus  $0 \preceq \mathcal{N}_A(Z) \preceq A$ , and  $\mathcal{N}_A(Z)$  is monotone in range  $Z$ : if  $\text{range } Z \subseteq \text{range } Z'$  then  $A - \mathcal{N}_A(Z') \preceq A - \mathcal{N}_A(Z)$ .

**Lemma 2.1** (Adding a block). *Let  $R = A - \mathcal{N}_A(Z)$  and let  $W$  be an additional real block. Then*

$$A - \mathcal{N}_A([Z \ W]) = R - RW(W^\top RW)^\dagger W^\top R. \quad (2.2)$$

*Proof.* Write  $P = P_Z$  and orthogonalize the new block against the old sketch:

$$H = (I - P)A^{1/2}W.$$

The enlarged sketch range is the orthogonal sum of  $\text{range}(A^{1/2}Z)$  and  $\text{range } H$ . Its projector is therefore

$$P_{[Z \ W]} = P + H(H^\top H)^\dagger H^\top.$$

The two identities

$$H^\top H = W^\top RW, \quad A^{1/2}H = RW$$

now turn the projector formula (2.1) into (2.2).  $\square$

Independently drawn blocks can therefore be analyzed as successive updates, conditioning on the current residual; the algorithm never inspects the residual before choosing its queries.

**Lemma 2.2** (Scalar lower bound for block progress). *For complex  $z$  and  $W = [\Re z \ \Im z]$ , and real  $R \succeq 0$ , the trace drop in (2.2) is at least*

$$\frac{z^* R^2 z}{z^* R z}, \quad (2.3)$$

*with the convention  $0/0 = 0$ . For a real one-column probe the drop equals this quotient.*

*Proof.* Let  $P$  project onto  $\text{range}(R^{1/2}W)$ ; the drop is  $\text{tr}(PR)$ . If  $z^* R z > 0$ , the complex unit vector  $v = R^{1/2}z/\sqrt{z^* R z}$  lies in the complexification of this real range, so  $P - vv^* \succeq 0$  and  $\text{tr}(PR) \geq v^* R v$ , which is (2.3). If  $z^* R z = 0$  both columns are annihilated by  $R^{1/2}$ .  $\square$

**Gaussian factors.** A standard Gaussian vector is a positive radius times an independent uniform direction, and multiplying a sketch column (or a whole complex block) by a nonzero scalar leaves (2.1) unchanged. Hence Gaussian Khatri–Rao columns have exactly the same Nyström output, under this coupling, as spherical product columns; every result below applies to them. With continuous factors, rank- $s$  inputs are recovered after  $s$  probes almost surely: product vectors span the space, so a suitable minor is a nonzero polynomial in the factor coordinates.

### 3 Product moments and the constant at order one

The probe is isotropic,  $\mathbb{E}zz^* = I$ . For  $p > -1/2$  (real) or  $p > -1$  (complex) put

$$c_n^{\mathbb{R}}(p) = \frac{n^p \Gamma(p+1/2) \Gamma(n/2)}{\Gamma(1/2) \Gamma(n/2+p)}, \quad c_n^{\mathbb{C}}(p) = \frac{n^p \Gamma(p+1) \Gamma(n)}{\Gamma(n+p)}, \quad (3.1)$$

and  $H(p) = \prod_j c_{n_j}(p)$ , the superscript matching the probe field. For  $n = 2$ :  $c_2^{\mathbb{C}}(p) = 2^p/(1+p)$ .

**Lemma 3.1** (Directional and quadratic moments). *Let the factors  $x_j$  be independent and isotropic, and suppose  $\mathbb{E}|u^*x_j|^{2p} \leq m_j(p)\|u\|^{2p}$  for all  $u$ , with  $p \geq 1$ . Then for every  $u$  and every Hermitian  $B \succeq 0$ ,*

$$\mathbb{E}|u^*z|^{2p} \leq M(p)\|u\|^{2p}, \quad \mathbb{E}(z^*Bz)^p \leq M(p)(\text{tr } B)^p, \quad M(p) = \prod_j m_j(p). \quad (3.2)$$

For spherical factors one may take  $m_j = c_{n_j}$ , with equality in the directional bound for a single factor, so  $M = H$ .

*Proof.* For one spherical factor and unit  $u$ , rotational invariance reduces  $|u^*x|^2$  to  $n$  times the squared first coordinate of a uniform unit vector, which is  $\text{Beta}(1/2, (n-1)/2)$  (real) or  $\text{Beta}(1, n-1)$  (complex); the beta moments give (3.1). For  $n = 1$  both equal one.

To tensorize, decompose  $u = \sum_a u_a \otimes e_a$  in the last factor. Conditioning on  $z_{<q} = x_1 \otimes \cdots \otimes x_{q-1}$ , the vector  $v_a = u_a^* z_{<q}$  is fixed and  $u^*z = \sum_a v_a(x_q)_a$ , so the local bound leaves  $m_q(p)(\sum_a |u_a^* z_{<q}|^2)^p$ . Minkowski in  $L^p$  gives

$$\left\| \sum_a |u_a^* z_{<q}|^2 \right\|_{L^p} \leq \sum_a (\mathbb{E}|u_a^* z_{<q}|^{2p})^{1/p} \leq \left( \prod_{j < q} m_j(p) \right)^{1/p} \sum_a \|u_a\|^2,$$

and induction proves the directional bound. For the quadratic bound write  $B = \sum_i b_i u_i u_i^*$  with unit  $u_i$  and apply Minkowski:  $\|z^*Bz\|_{L^p} \leq \sum_i b_i \| |u_i^* z|^2 \|_{L^p} \leq M(p)^{1/p} \text{tr } B$ .  $\square$

**Lemma 3.2** (The order-one constant). *Each  $p \mapsto \log m(p)$  that is a supremum of logarithmic moments  $\log \mathbb{E}|u^*x|^{2p}$  over unit  $u$  is convex, with  $m(1) = 1$  under isotropy. Consequently  $\beta \mapsto M(1+\beta)^{1/\beta}$  is nondecreasing on  $(0, \infty)$  and*

$$\kappa := \lim_{\beta \downarrow 0} M(1+\beta)^{1/\beta} = \inf_{\beta > 0} M(1+\beta)^{1/\beta} \leq M(2).$$

For spherical factors  $\kappa = \kappa_{\mathbb{R}}$  or  $\kappa_{\mathbb{C}}$  of (1.2)–(1.3), and  $D_n < \log c_n(2)$  for  $n \geq 2$ .

*Proof.*  $p \mapsto \log \mathbb{E}X^p$  is convex by Hölder, and a supremum of convex functions is convex. A convex function vanishing at 0 has nondecreasing secant slopes  $\beta \mapsto \log M(1+\beta)/\beta$ , whose limit at 0 is the right derivative. For spheres, differentiate (3.1) at  $p = 1$ : in the complex case  $\log n + \psi(2) - \psi(n+1) = \log n + 1 - H_n$  using  $\psi(k+1) = H_k - \gamma$ ; in the real case  $\log n + \psi(3/2) - \psi(n/2+1)$ . Strict convexity for  $n \geq 2$  gives  $D_n < \log c_n(2) - \log c_n(1)$ .  $\square$

Since  $\mathbb{E}X = 1$  for  $X = |u^*x|^2$  with unit  $u$ , the slope is  $\mathbb{E}[X \log X] \geq 0$ :  $\log \kappa$  is a sum of size-biased logarithmic moments, and  $\kappa \geq 1$ .

| Local law            | Order-one constant $e^{D_n}$                                   | Fourth moment $c_n(2)$ |
|----------------------|----------------------------------------------------------------|------------------------|
| Real, $n = 2$        | $e/2 = 1.359141$                                               | $3/2$                  |
| Complex, $n = 2$     | $2/\sqrt{e} = 1.213061$                                        | $4/3$                  |
| Real, general $n$    | $e^{\log n + \psi(3/2) - \psi(n/2+1)} \uparrow e^{2-\gamma}/2$ | $3n/(n+2)$             |
| Complex, general $n$ | $e^{\log n + 1 - H_n} \uparrow e^{1-\gamma}$                   | $2n/(n+1)$             |

Improving each factor gives an exponential improvement in  $q$ .

## 4 From fractional moments to expected progress

The numerator and denominator of (2.3) depend on the same probe. A change of measure by the numerator keeps this dependence and reduces the problem to a mixed moment.

**Lemma 4.1** (Mixed moment and ratio conversion). *Let  $z$  be isotropic and satisfy the directional bound of (3.2) at  $p = 1 + \beta$ ,  $\beta > 0$ . For nonzero PSD  $B, R$  with  $\ker R \subseteq \ker B$ ,*

$$\mathbb{E} \frac{z^* B z}{z^* R z} \geq \frac{\operatorname{tr} B}{M(1 + \beta)^{1/\beta} \operatorname{tr} R}. \quad (4.1)$$

*Proof.* Set  $a = z^* B z$ ,  $b = z^* R z$ , and  $p = 1 + \beta$ . The kernel condition ensures  $a = 0$  on  $\{b = 0\}$ , and isotropy gives  $\mathbb{E} a = \operatorname{tr} B$ .

*Step 1: weight by the numerator.* Define the probability measure

$$dQ = \frac{a}{\operatorname{tr} B} dP.$$

Under  $Q$ , the denominator is positive almost surely. Jensen's inequality for  $t \mapsto t^{-1/\beta}$  gives

$$\mathbb{E} \frac{a}{b} = \operatorname{tr} B \mathbb{E}_Q b^{-1} \geq \operatorname{tr} B (\mathbb{E}_Q b^\beta)^{-1/\beta}. \quad (4.2)$$

The conclusion also holds when the left side is infinite.

*Step 2: bound the mixed moment.* Take a spectral decomposition  $B = \sum_i b_i u_i u_i^*$  with unit eigenvectors. Hölder with conjugate exponents  $p$  and  $p/\beta$ , followed by Lemma 3.1, yields

$$\begin{aligned} \mathbb{E}[|u_i^* z|^2 b^\beta] &\leq (\mathbb{E}|u_i^* z|^{2p})^{1/p} (\mathbb{E} b^p)^{\beta/p} \\ &\leq M(p)^{1/p} M(p)^{\beta/p} (\operatorname{tr} R)^\beta \\ &= M(p) (\operatorname{tr} R)^\beta. \end{aligned}$$

Summing with weights  $b_i$ , we obtain

$$\mathbb{E}_Q b^\beta = \frac{\mathbb{E}[ab^\beta]}{\operatorname{tr} B} \leq M(p) (\operatorname{tr} R)^\beta.$$

Substitution into (4.2) proves (4.1). □

**Theorem 4.2** (Order-one drift). *Under the hypotheses of Lemma 3.1 for all  $p \in (1, 1 + \beta_0)$ , and with  $\kappa$  as in Lemma 3.2, one fresh probe or complex block satisfies  $0 \leq R' \preceq R$  and*

$$\mathbb{E}(\operatorname{tr} R - \operatorname{tr} R') \geq \frac{\operatorname{tr}(R^2)}{\kappa \operatorname{tr} R}. \quad (4.3)$$

More generally, for  $B \succeq 0$  with  $\ker R \subseteq \ker B$ ,

$$\mathbb{E} \frac{z^* B z}{z^* R z} \geq \frac{\operatorname{tr} B}{\kappa \operatorname{tr} R}. \quad (4.4)$$

*Proof.* Take  $B = R^2$  in Lemma 4.1, bound the drop below by the quotient (Lemma 2.2), and let  $\beta \downarrow 0$ ; the left side does not depend on  $\beta$ . The projector representation gives  $0 \preceq R' \preceq R$ .  $\square$

No lower spectral edge of a sample Gram matrix and no inverse moment are needed. Taking  $\beta = 1$  gives the fourth-moment constant  $M(2)$ ; letting  $\beta \rightarrow 0$  improves it without changing the algorithm.

## 5 Unconditional Kronecker Nyström approximation

Let  $R_t = A - \widehat{A}_t$  after  $t$  independent probes or blocks and  $T = \operatorname{tr} A$ . If  $T = 0$  then  $A = 0$  and everything is trivial; otherwise  $R_t$  decreases in PSD order and  $\operatorname{tr} R_t \leq T$ . All statements in this section hold for every fixed real PSD  $A$ , whatever its eigenvectors, and for any probe law satisfying Theorem 4.2 (in particular Gaussian Khatri–Rao columns with  $\kappa = \kappa_{\mathbb{R}}$ ).

### 5.1 Frobenius residual

**Proposition 5.1.**  $\mathbb{E} \operatorname{tr}(R_m^2) \leq \kappa T^2/m$  for every  $m \geq 1$ .

*Proof.* Conditional drift gives  $\mathbb{E}[\operatorname{tr} R_t - \operatorname{tr} R_{t+1} \mid R_t] \geq \operatorname{tr}(R_t^2)/(\kappa T)$ . Summing from  $t = 0$  to  $m-1$  telescopes to  $\sum_{t < m} \mathbb{E} \operatorname{tr}(R_t^2) \leq \kappa T^2$ . If  $0 \preceq S \preceq R$  then  $\operatorname{tr} R^2 - \operatorname{tr} S^2 = \operatorname{tr}((R-S)(R+S)) \geq 0$ , so the summands decrease and the sum is at least  $m \mathbb{E} \operatorname{tr}(R_m^2)$ .  $\square$

Section 5.3 sharpens this to  $\kappa T^2/(m + \kappa T^2/\|A\|_F^2)$ .

### 5.2 The $(1 + \varepsilon)$ trace-tail theorem

**Theorem 5.2** (Unconditional Kronecker Nyström, relative trace error). *Let  $1 \leq r < \operatorname{rank} A$ ,  $\tau = \tau_r(A) > 0$  and  $0 < \varepsilon \leq 1$ . If*

$$m \geq \left\lceil \kappa r \left[ \frac{1}{\varepsilon} + \log \frac{1}{\varepsilon} + \log \frac{T}{\tau} \right] \right\rceil, \quad (5.1)$$

*then  $\mathbb{E} \operatorname{tr} R_m \leq (1 + \varepsilon)\tau$ . The approximation is not truncated to rank  $r$ .*

*Proof. Step 1: reduce the spectrum to the residual trace.* Since  $R_t \preceq A$ , its rank- $r$  tail is at most  $\tau$ . The first  $r$  eigenvalues therefore sum to at least  $(\operatorname{tr} R_t - \tau)_+$ . Cauchy–Schwarz implies

$$\operatorname{tr} R_t^2 \geq \frac{(\operatorname{tr} R_t - \tau)_+^2}{r}.$$

The function  $g(x) = (x - \tau)_+^2/x$ , extended by  $g(0) = 0$ , is convex. Above  $\tau$ , it equals  $x - 2\tau + \tau^2/x$ ; at  $\tau$  its derivative matches the zero derivative on the interval below.

*Step 2: obtain a scalar recurrence.* Let  $f_t = \mathbb{E} \operatorname{tr} R_t$ . Conditional drift and Jensen give

$$f_{t+1} \leq f_t - \frac{(f_t - \tau)_+^2}{\kappa r f_t}, \quad (5.2)$$

where a zero denominator is interpreted by continuity. For  $u_t = f_t/\tau - 1 > 0$ , this becomes

$$u_t - u_{t+1} \geq \frac{u_t^2}{\kappa r(1 + u_t)}.$$

*Step 3: count the steps until the target is reached.* The decreasing convex function

$$F(u) = u^{-1} - \log u, \quad F'(u) = -\frac{1+u}{u^2}$$

satisfies, whenever  $u_{t+1} > 0$ ,

$$F(u_{t+1}) - F(u_t) \geq F'(u_t)(u_{t+1} - u_t) \geq \frac{1}{\kappa r}.$$

If  $u_s > \varepsilon$  for all  $0 \leq s \leq t$ , telescoping gives

$$F(u_t) \geq F(u_0) + \frac{t}{\kappa r} \geq -\log(T/\tau) + \frac{t}{\kappa r}.$$

For the integer  $t$  in (5.1), the right side is at least  $F(\varepsilon) = \varepsilon^{-1} + \log(\varepsilon^{-1})$ . This contradicts  $u_t > \varepsilon$ , because  $F$  is decreasing. Thus some  $s \leq t$  satisfies  $f_s \leq (1 + \varepsilon)\tau$ , and monotonicity of the residual trace preserves the bound at every later step.  $\square$

This is the scalar residual-contraction mechanism of RPCholesky analysis [5] with the product law charged through  $\kappa$ . It applies to the full (untruncated) Nyström output. For Gaussian Khatri–Rao sketches of order  $q$  with  $n_j \rightarrow \infty$  the constant tends to  $\kappa_0^q$ ,  $\kappa_0 = 2.0743$ ; e.g.  $\kappa_0^2 = 4.30$  at  $q = 2$ . Unlike results that assume a common Kronecker (Schmidt) structure of the eigenvectors of  $A$ , or that rest on Gaussian equivalence, no hypothesis ties the eigenvectors of  $A$  to the tensor factors; the price is  $\kappa$  and the logarithm of  $T/\tau_r$ .

### 5.3 Convex potentials and Schatten rates

**Theorem 5.3** (Convex-potential progress). *Let  $f : [0, \infty) \rightarrow [0, \infty)$  be finite and convex with  $f(0) = 0$ . For nonzero real  $R \succeq 0$ , one fresh probe or block gives*

$$\mathbb{E} \operatorname{tr} f(R') \leq \operatorname{tr} f(R) - \frac{\operatorname{tr}[Rf(R)]}{\kappa \operatorname{tr} R}. \quad (5.3)$$

*Proof.* Let  $P$  be the projector onto  $\operatorname{range}(R^{1/2}W)$ , and set  $Q = I - P$ . The matrices  $R' = R^{1/2}QR^{1/2}$  and  $QRQ$  have the same eigenvalues, including multiplicities of zero.

*Compression.* Choose an orthonormal eigenbasis  $v_a$  for the compression of  $R$  to  $\operatorname{range} Q$ . Scalar Jensen applied to the spectral measure of  $R$  in the state  $v_a$  gives

$$f(v_a^* R v_a) \leq v_a^* f(R) v_a.$$

Summing and using  $f(0) = 0$ , we conclude

$$\operatorname{tr} f(R') \leq \operatorname{tr}(Qf(R)) = \operatorname{tr} f(R) - \operatorname{tr}(Pf(R)).$$

*Expected loss of potential.* When  $z^* R z > 0$ , the unit vector  $v = R^{1/2}z/\sqrt{z^* R z}$  lies in the complexification of the range of  $P$ . Since  $f(R) \succeq 0$ ,

$$\operatorname{tr}(Pf(R)) \geq v^* f(R) v = \frac{z^* R f(R) z}{z^* R z}.$$

At a zero denominator, use the zero-quotient convention. Apply (4.4) with  $B = Rf(R)$ , which is PSD and vanishes on  $\ker R$ . This proves the desired bound. The compression step is the argument of [6, Lemma 4.2].  $\square$

**Theorem 5.4** (Schatten residual decay). *For every real  $p > 1$  and  $m \geq 0$ ,*

$$\begin{aligned} \mathbb{E} \operatorname{tr}(R_m^p) &\leq T^p \left[ \left( \frac{T^p}{\operatorname{tr} A^p} \right)^{1/(p-1)} + \frac{m}{\kappa(p-1)} \right]^{-(p-1)} \\ &\leq \frac{T^p}{(1 + m/[\kappa(p-1)])^{p-1}}. \end{aligned} \quad (5.4)$$

In particular  $\mathbb{E} \|R_m\|_F^2 \leq \kappa T^2 / (m + \kappa T^2 / \|A\|_F^2) \leq \kappa T^2 / (m + \kappa)$ .

*Proof. Moment interpolation.* Hölder's inequality for the eigenvalues gives

$$\operatorname{tr} R^p \leq (\operatorname{tr} R)^{1/p} (\operatorname{tr} R^{p+1})^{(p-1)/p},$$

and consequently

$$\operatorname{tr} R^{p+1} \geq (\operatorname{tr} R)^{p/(p-1)} (\operatorname{tr} R)^{-1/(p-1)}.$$

Use  $f(x) = x^p$  in Theorem 5.3 and  $\operatorname{tr} R_t \leq T$  to obtain

$$\mathbb{E}[\operatorname{tr} R_{t+1}^p \mid R_t] \leq \operatorname{tr} R_t^p - \frac{(\operatorname{tr} R_t^p)^{p/(p-1)}}{\kappa T^{p/(p-1)}}.$$

*Solve the recurrence.* Set  $a_t = \mathbb{E} \operatorname{tr} R_t^p / T^p$ . Jensen now implies

$$a_{t+1} \leq a_t - \kappa^{-1} a_t^{p/(p-1)}.$$

For  $\phi(x) = x^{-1/(p-1)}$ , convexity and monotonicity give, as long as  $a_{t+1} > 0$ ,

$$\phi(a_{t+1}) - \phi(a_t) \geq \phi'(a_t)(a_{t+1} - a_t) \geq \frac{1}{\kappa(p-1)}.$$

Summing and inverting proves the first inequality in (5.4). If any  $a_t$  vanishes, the residual remains zero and the conclusion is immediate. Finally  $\operatorname{tr} A^p \leq T^p$  gives the second inequality.  $\square$

## 5.4 Operator-norm tail and a regularized preconditioner

**Theorem 5.5** (Absolute spectral error). *For  $0 < \mu \leq T$ ,*

$$\mathbb{P}\{\|R_m\| \geq \mu\} \leq \min\left\{1, \frac{2T}{\mu} \exp\left(-\frac{m\mu}{2\kappa T}\right)\right\}. \quad (5.5)$$

*Consequently, for  $0 < \rho \leq T$  and  $0 < \delta < 1$ , if  $m \geq (2\kappa T/\rho) \log(2T/(\rho\delta))$  then with probability at least  $1 - \delta$  the matrix  $P = \hat{A}_m + \rho I$  satisfies  $P \preceq A + \rho I \preceq 2P$ , so  $\operatorname{cond}(P^{-1/2}(A + \rho I)P^{-1/2}) \leq 2$ .*

*Proof.* Use the ramp  $f(x) = (x - \mu/2)_+$ , for which  $xf(x) \geq (\mu/2)f(x)$ . By (5.3) and  $\operatorname{tr} R_t \leq T$ ,  $\mathbb{E}[\operatorname{tr} f(R_{t+1}) \mid R_t] \leq (1 - \mu/(2\kappa T)) \operatorname{tr} f(R_t)$ , a multiplier in  $[0, 1]$  since  $\kappa \geq 1$ . Iterate from  $\operatorname{tr} f(A) \leq T$  and use  $1 - u \leq e^{-u}$ . On  $\{\|R_m\| \geq \mu\}$  the potential is at least  $\mu/2$ ; apply Markov. For the preconditioner, on  $\{\|R_m\| < \rho\}$  we have  $0 \preceq R_m \preceq \rho I \preceq P$  and  $A + \rho I = P + R_m$ .  $\square$

The construction uses only product queries; Nyström preconditioning itself is standard [11]. The bound is an absolute (trace-scale) statement, not a relative rank- $r$  spectral theorem.

## 6 A trace estimator using legal real queries

### 6.1 Complex responses from real product queries

Write the complex factors as  $a_j + ib_j$  and form  $v(t) = \bigotimes_j (a_j + tb_j)$ , a vector polynomial of degree at most  $q$ . Query  $Av(t_k)$  at  $q + 1$  distinct real nodes and interpolate at  $t = i$ ; this yields  $Az$ , hence  $AW$ . Every node vector is a legal real product [4, Lemma 4.2]. Raw Gaussian factors suffice, since their radial scale cancels from the block range.

**Proposition 6.1** (Retaining the interpolation queries). *Let the stage-one sketch contain all  $q + 1$  real vectors  $v(t_k)$  of each block, instead of only  $W = [\Re z \ \Im z]$ . Then its residual is pathwise PSD-dominated by the residual of the block sketch, and Theorems 4.2–5.5 and 6.4 hold verbatim for it.*

*Proof.*  $z = v(i) = \sum_k \ell_k(i)v(t_k)$  with Lagrange weights, so  $\Re z, \Im z \in \text{range}_{\mathbb{R}}[v(t_0), \dots, v(t_q)]$ . Nyström residuals are monotone in the sketch range (Section 2), so each step’s drop is at least the block’s drop, and the retained update of step  $t$  is independent of earlier steps. All proofs above use only the lower bound on the conditional drop and PSD monotonicity.  $\square$

Retention costs no additional calls and is the default. The lower bound in Section 8 concerns only the two-column block. The larger retained span is not covered by that lower bound; its asymptotic advantage remains open.

### 6.2 Training and correcting the residual diagonal

Condition on stage one; write  $C = \hat{A}_m$ ,  $R = A - C$ ,  $d_i = R_{ii}$ . From  $Y = AZ$ ,  $C = Y(Z^\top Y)^\dagger Y^\top$  is known, so  $Rx = Ax - Cx$  costs one call. Draw  $s$  independent product-Rademacher vectors  $x^{(a)}$  (each a tensor product of Rademacher factors); they satisfy  $(x_i^{(a)})^2 = 1$  and  $\mathbb{E}x_i^{(a)}x_j^{(a)} = \mathbf{1}_{i=j}$ . Put  $h_i = s^{-1} \sum_a x_i^{(a)}(Rx^{(a)})_i$ .

**Lemma 6.2** (Diagonal training).  $\mathbb{E}h = d$  and  $\mathbb{E}\|h - d\|^2 = (\|R\|_F^2 - \sum_i d_i^2)/s$ .

*Proof.* One sample has error  $\sum_{j \neq i} R_{ij}x_i x_j$  in coordinate  $i$ ; squaring,  $x_i^2 = 1$  and isotropy leave  $\sum_{j \neq i} R_{ij}^2$ . Average over  $s$  samples and sum.  $\square$

Choose known probabilities  $\pi_i \geq 1/(2N)$  (e.g. a uniform  $\lceil \log_2 N \rceil$ -bit integer reduced modulo  $N$ ), draw  $s$  independent indices  $I_\ell$ , query the coordinate products  $e_{I_\ell}$ , and return

$$\hat{T} = \text{tr } C + \sum_i h_i + \frac{1}{s} \sum_{\ell=1}^s \frac{d_{I_\ell} - h_{I_\ell}}{\pi_{I_\ell}}. \quad (6.1)$$

This is the diagonal-control construction of [4, Section 7] applied to the Nyström residual.

**Lemma 6.3.**  $\mathbb{E}[\hat{T} \mid R, h] = \text{tr } A$  and  $\text{Var}(\hat{T} \mid R) \leq 2N\|R\|_F^2/s^2$ .

*Proof.* Given  $R, h$  the correction is an average of  $s$  i.i.d. terms with mean  $\sum_i (d_i - h_i)$  and second moment  $\sum_i (d_i - h_i)^2/\pi_i \leq 2N\|h - d\|^2$ . The conditional mean is  $\text{tr } A$  for every  $h$ , so there is no variance of a conditional mean. Apply Lemma 6.2.  $\square$

### 6.3 The query bound

**Theorem 6.4** (Trace estimation with real product queries). *Fix  $(b, \kappa) = (1, e^{qD_n^{\mathbb{R}}})$  (real probes) or  $(q+1, e^{qD_n^{\mathbb{C}}})$  (complex probes implemented over  $\mathbb{R}$ ). For integers  $m, s \geq 1$  the estimator (6.1) is unbiased, uses  $bm + 2s$  real product queries, all chosen before any response is seen, and*

$$\text{Var } \hat{T} \leq \frac{2N\kappa}{ms^2} T^2. \quad (6.2)$$

With  $L_b = (6bN\kappa/\varepsilon^2)^{1/3}$ ,  $m = \lceil L_b/b \rceil$ ,  $s = \lceil L_b \rceil$  it succeeds with probability at least  $2/3$  using  $\mathcal{D}_b := \min\{N, b\lceil L_b/b \rceil + 2\lceil L_b \rceil\} \leq \min\{N, 3L_b + b + 2\}$  queries. The estimator runs the branch with the smaller  $\mathcal{D}_b$ , a choice depending only on  $(n, q, \varepsilon)$ , and so uses  $\min\{\mathcal{D}_1, \mathcal{D}_{q+1}\}$  queries. For every  $\eta > 0$  a version with a bounded number of fair random bits satisfies the same with  $\kappa$  replaced by  $(1 + \eta)\kappa$  (Appendix A).

*Proof. Variance.* Condition on the first-stage sketch and use Lemma 6.3, then Proposition 5.1:

$$\text{Var } \hat{T} \leq \frac{2N}{s^2} \mathbb{E} \|R_m\|_F^2 \leq \frac{2N\kappa}{ms^2} T^2.$$

Chebyshev's inequality therefore gives

$$\mathbb{P}\{|\hat{T} - T| > \varepsilon T\} \leq \frac{2N\kappa}{ms^2\varepsilon^2}.$$

With the stated integers,  $ms^2 \geq L_b^3/b = 6N\kappa/\varepsilon^2$ , so this probability is at most  $1/3$ .

*Calls and nonadaptivity.* Stage one costs  $bm$  calls, diagonal training costs  $s$ , and coordinate correction costs another  $s$ . All three query lists can be sampled in advance; only the arithmetic combining their responses is sequential. Rounding gives

$$b\lceil L_b/b \rceil + 2\lceil L_b \rceil \leq 3L_b + b + 2.$$

Alternatively, a coordinate-basis sweep computes the trace exactly in  $N$  legal calls. Choosing the cheaper branch uses only the known parameters, so it preserves unbiasedness and nonadaptivity. The finite-law statement follows from Proposition A.2.  $\square$

**Proposition 6.5** (Which branch wins at  $n = 2$ ). *At  $n = 2$  the leading terms compare as  $6N(e/2)^q$  (real) versus  $6N(q+1)(2/\sqrt{e})^q$  (complex). The real branch has the smaller leading term if and only if  $(e^{3/2}/4)^q \leq q+1$ , i.e. exactly for  $q \leq 30$ .*

*Proof.* The ratio is  $(e^{3/2}/4)^q/(q+1)$  with  $e^{3/2}/4 = 1.120422$ . The sequence  $\rho_q = 1.120422^q/(q+1)$  decreases while  $q+2 > 1.120422(q+1)$ , i.e. for  $q \leq 7$ , and increases afterwards;  $\rho_1 < 1$ ,  $\rho_{30} = 30.30/31 < 1$  and  $\rho_{31} = 33.95/32 > 1$ .  $\square$

With the exact integer budgets the same crossover  $q = 30$  holds at  $\varepsilon = (9/10)^q$  and at  $\varepsilon = 0.1$  for all  $2 \leq q \leq 60$  (Section 10).

## 7 Comparison with explicit benchmark budgets

This section compares the new bound with two explicit budget formulas recorded in the source draft's earlier head-tail analysis. We define both formulas here; the comparisons below are algebraic statements about these budgets and do not require the earlier report or its proofs. They

do not claim domination of every published algorithm. Write  $b \in \{1, q+1\}$  and  $\kappa_4 = c_n(2)^q$ , i.e.  $(3n/(n+2))^q$  (real) or  $(2n/(n+1))^q$  (complex).

- *Fourth-moment branch* (an Oliveira lower edge plus a fourth moment):

$$\mathcal{U}_1^{(b)} = \min\left\{N, \min_{1 \leq r < N} \left( b \lceil 196\kappa_4(r + 2 \log 24) \rceil + 2 \left\lceil \sqrt{192N/(r\varepsilon^2)} \right\rceil \right) \right\}.$$

- *Fractional joint-cap branch* (complex blocks), with  $b_p = (52^p c_n^{\mathbb{C}}(p)^q)^{1/(p-1)}$ :

$$\mathcal{U}_2 = \min\left\{N, \inf_{1 < p \leq 2} \min_{1 \leq r < N} \left( (q+1) \lceil b_p(r + 2 \log 24) \rceil + 2 \left\lceil \sqrt{192N/(r\varepsilon^2)} \right\rceil \right) \right\}.$$

**Theorem 7.1** (Domination). *For every  $n, q \geq 2$  and  $0 < \varepsilon < 1/2$ :  $\mathcal{D}_1 \leq \mathcal{U}_1^{(1)}$ ,  $\mathcal{D}_{q+1} \leq \mathcal{U}_1^{(q+1)}$  and  $\mathcal{D}_{q+1} \leq \mathcal{U}_2$ . Before capping,*

$$\begin{aligned} (\mathcal{U}_1^{(b)})^{\text{uncapped}} &\geq (196 \cdot 32)^{1/3} 3L_b + 392 b \log 24, \\ \mathcal{U}_2^{\text{uncapped}} &\geq (2704 \cdot 32)^{1/3} 3L_{q+1} + 5408 (q+1) \log 24. \end{aligned}$$

The multiplicative constants are approximately 18.44 and 44.23.

*Proof.* We compare the formulas before applying the common cap  $N$ .

*Step 1: minimize the head–tail tradeoff.* For  $\alpha, \beta > 0$  and  $r > 0$ , elementary minimization gives

$$\alpha r + \beta r^{-1/2} \geq 3(\alpha\beta^2/4)^{1/3}.$$

This bound holds for every admissible integer  $r$ , and dropping ceilings can only lower the earlier budget.

*Step 2: the fourth-moment budget.* Take

$$\alpha = 196b\kappa_4, \quad \beta = \frac{2\sqrt{192N}}{\varepsilon}.$$

The uncapped fourth-moment expression is at least

$$3(196 \cdot 192 b\kappa_4 N/\varepsilon^2)^{1/3} + 392b\kappa_4 \log 24.$$

Lemma 3.2 gives  $\kappa \leq \kappa_4$ , so its first term is at least

$$(196 \cdot 32)^{1/3} 3L_b.$$

The remaining term is larger than  $b + 2$ , since  $\kappa_4 \geq 1$ . The entire expression therefore exceeds  $3L_b + b + 2$ , an upper bound for the rounded drift budget.

*Step 3: the fractional budget.* For  $1 < p \leq 2$ , convexity at order one and  $p/(p-1) \geq 2$  imply

$$\begin{aligned} c_n^{\mathbb{C}}(p)^{1/(p-1)} &\geq e^{D_n^{\mathbb{C}}}, \\ b_p &= 52^{p/(p-1)} c_n^{\mathbb{C}}(p)^{q/(p-1)} \geq 2704\kappa_{\mathbb{C}}. \end{aligned}$$

Repeat Step 1 with  $\alpha = (q+1)b_p$ . The resulting lower bound is

$$(2704 \cdot 32)^{1/3} 3L_{q+1} + 5408(q+1) \log 24.$$

Again the additive term absorbs the rounding allowance. These inequalities hold uniformly in  $p$  and  $r$ , hence also after their infimum and minimum. Finally, applying  $x \mapsto \min\{N, x\}$  preserves each comparison.  $\square$

**Proposition 7.2** (The joint-cap overhead is genuinely  $\exp(\Theta(\sqrt{q}))$ ). *At  $n = 2$ , for every  $q \geq 1$ ,*

$$52 \exp\left\{qD_2^{\mathbb{C}} + \sqrt{\frac{2 \log 52}{9} q}\right\} \leq \inf_{1 < p \leq 2} b_p,$$

*with  $\sqrt{2 \log 52/9} = 0.9370$ , and for  $q \geq 8 \log 52$  also  $\inf_{1 < p \leq 2} b_p \leq 52 \exp\{qD_2^{\mathbb{C}} + \sqrt{(\log 52/2) q}\}$ , with  $\sqrt{\log 52/2} = 1.4056$ . Consequently, before capping,*

$$\mathcal{U}_2^{\text{uncapped}} \geq (52 \cdot 32)^{1/3} e^{0.3123\sqrt{q}} \cdot 3L_{q+1} \geq 11.84 e^{0.3123\sqrt{q}} (\mathcal{D}_{q+1} - q - 3),$$

*while  $\mathcal{D}_{q+1} \leq 3(6(q+1))^{1/3}(\kappa_{\mathbb{C}}N/\varepsilon^2)^{1/3} + q + 3$  carries only a polynomial prefactor.*

*Proof.* Put  $t = p - 1 \in (0, 1]$  and  $a = \log 52$ . For the complex binary moment function,

$$(\log c_2)''(p) = (1 + p)^{-2} \in [1/9, 1/4] \quad (1 \leq p \leq 2).$$

Taylor's formula at  $p = 1$  therefore yields

$$D_2^{\mathbb{C}}t + \frac{t^2}{18} \leq \log c_2^{\mathbb{C}}(1 + t) \leq D_2^{\mathbb{C}}t + \frac{t^2}{8}.$$

*Lower bound.* The definition of  $b_p$  gives

$$\begin{aligned} \log b_{1+t} &= (1 + t^{-1})a + t^{-1}q \log c_2^{\mathbb{C}}(1 + t) \\ &\geq a + qD_2^{\mathbb{C}} + \frac{a}{t} + \frac{qt}{18} \\ &\geq a + qD_2^{\mathbb{C}} + \sqrt{\frac{2aq}{9}}. \end{aligned}$$

The last line is AM–GM, so it remains valid after taking the infimum in  $t$ .

*Matching upper order.* If  $q \geq 8a$ , choose  $t = \sqrt{8a/q} \leq 1$ . The upper Taylor bound gives

$$\log b_{1+t} \leq a + qD_2^{\mathbb{C}} + \sqrt{\frac{aq}{2}}.$$

This proves the stated  $\exp(\Theta(\sqrt{q}))$  overhead in the coefficient  $b_p$ , relative to  $\kappa_{\mathbb{C}}$ .

*Translate to calls.* Step 1 of Theorem 7.1, now with  $\alpha = (q + 1)b_p$ , gives

$$\mathcal{U}_2^{\text{uncapped}} \geq 3((q + 1)b_p 192N/\varepsilon^2)^{1/3}.$$

Divide by  $3L_{q+1}$ , insert the lower coefficient bound, and use

$$(52 \cdot 192/6)^{1/3} = (52 \cdot 32)^{1/3}, \quad \frac{1}{3}\sqrt{2 \log 52/9} > 0.3123.$$

Lastly  $3L_{q+1} \geq \mathcal{D}_{q+1} - q - 3$ . The statements concern uncapped expressions; the exact basis cap may erase their separation.  $\square$

*Remark 7.3* (Binary running comparison). At  $n = 2$ ,  $\varepsilon = (9/10)^q$ ,  $\kappa N/\varepsilon^2 = (2e^D \cdot 100/81)^q$  and the bases are:

| Branch                                                       | Exponential base                      | Prefactor                     |
|--------------------------------------------------------------|---------------------------------------|-------------------------------|
| Direct diagonal control $2\lceil\sqrt{6N}/\varepsilon\rceil$ | $10\sqrt{2}/9 = 1.571348$             | $2\sqrt{6}$                   |
| Fourth moment, real                                          | $(100/27)^{1/3} = 1.547196$           | $3(196 \cdot 192)^{1/3}$      |
| Fourth moment, complex block                                 | $(800/243)^{1/3} = 1.487629$          | $3(196 \cdot 192(q+1))^{1/3}$ |
| Fractional joint cap                                         | $(400/(81\sqrt{e}))^{1/3} = 1.441482$ | $\exp(\Theta(\sqrt{q}))$      |
| <b>Drift, real</b>                                           | $(100e/81)^{1/3} = 1.497166$          | $3 \cdot 6^{1/3}$             |
| <b>Drift, complex block</b>                                  | $(400/(81\sqrt{e}))^{1/3} = 1.441482$ | $3(6(q+1))^{1/3}$             |
| Drift at conjectural $\kappa = e^{q^{7/2}}$ (complex)        | $(200e^{7/2}/81)^{1/3} = 1.378748$    | –                             |
| Analysis benchmark ( $\kappa$ subexponential)                | $(200/81)^{1/3} = 1.351600$           | –                             |

These are sufficient bounds for specified constructions. The comparison does not establish minimax exponents for the unrestricted oracle.

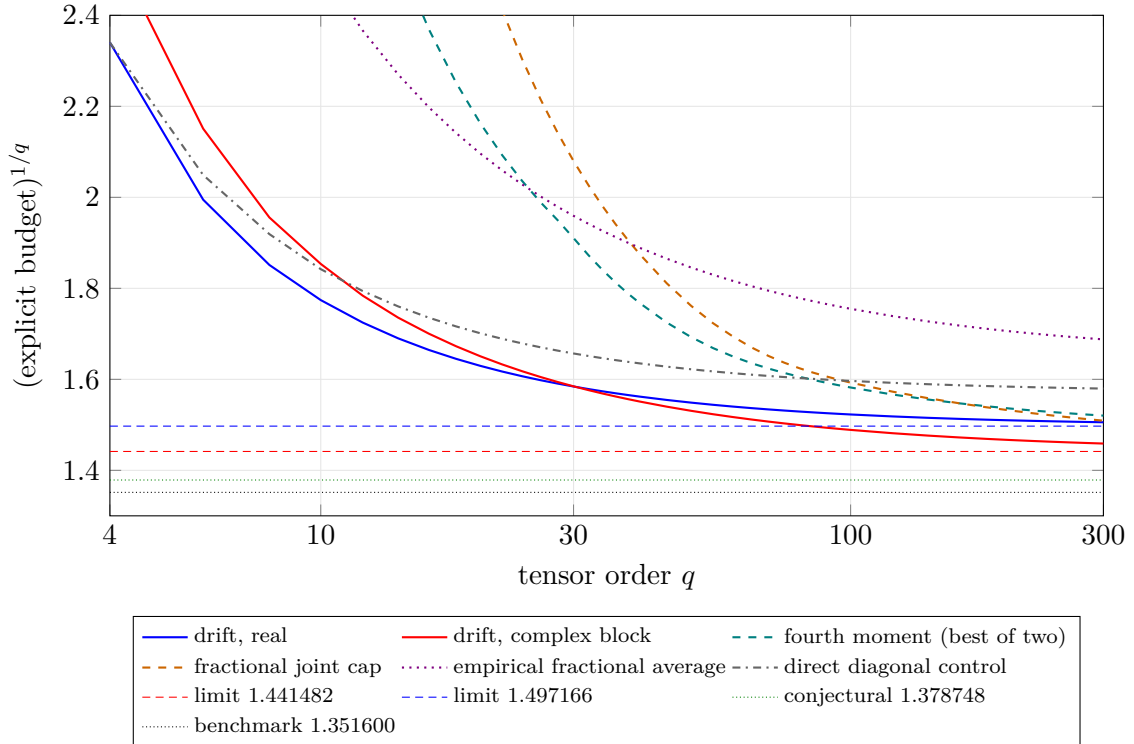


Figure 2: Effective upper bases (explicit uncapped budget)<sup>1/q</sup> against  $q$  at  $n = 2$ ,  $\varepsilon = (9/10)^q$ , computed from the exact integer budgets (drift branches, the fourth-moment and joint-cap benchmarks defined above, the empirical fractional average and direct diagonal control of [4]). Horizontal lines: the limiting drift bases, the conjectural base under Conjecture 8.4, and the analysis benchmark. The joint-cap curve exceeds its base 1.441482 by a factor of at least  $e^{0.31/\sqrt{q}}$  (Proposition 7.2), so it approaches the base only slowly.

**Complementary guarantees.** The drift estimator does not dominate every earlier bound. At fixed  $\varepsilon$ , Holden’s empirical fractional branch  $(q+1) \inf_{1 < p \leq 2} \lceil (6c_n(p)^q/\varepsilon^p)^{1/(p-1)} \rceil$  has base

$2/\sqrt{e} = 1.2131$  at  $n = 2$ , better than  $(4/\sqrt{e})^{1/3} = 1.3437$  (complex drift) and  $e^{1/3} = 1.3956$  (real drift); at  $\varepsilon = (9/10)^q$  its base is 1.6461 (attained at  $p = 2$ ). The combined sufficient upper bound is therefore

$$\min\{N, \text{ empirical fractional}, \mathcal{D}_1, \mathcal{D}_{q+1}\};$$

the earlier Nyström benchmark budgets are dominated by Theorem 7.1. Direct diagonal control remains an additional available bound, especially when low-rank approximation is inconvenient.

## 8 The sharp exponent

Theorem 4.2 is sufficient, not sharp. We now show that the per-factor exponent of the optimal drift constant is pinned between an explicit Chernoff rate and  $D_n$ , and that the two-column block does not change the lower end.

**Definition 8.1.** For the real one-column law or the complex block law on  $(\mathbb{R}^n)^{\otimes q}$  let

$$\kappa_{\text{opt}}(n, q) = \sup_{R \succeq 0, R \neq 0} \frac{\text{tr}(R^2)/\text{tr } R}{\mathbb{E}(\text{tr } R - \text{tr } R')},$$

the smallest constant for which (4.3) holds. Define

$$\gamma_n = - \min_{0 \leq \lambda \leq 1} \log c_n(\lambda)$$

with  $c_n = c_n^{\mathbb{R}}$  or  $c_n^{\mathbb{C}}$  matching the law.

Here  $c_n(\lambda) = \mathbb{E}X^\lambda$  for  $X = n|w_1|^2$ ,  $w$  uniform on the unit sphere, so  $e^{-q\gamma_n} = \min_\lambda \mathbb{E} \prod_j X_j^\lambda$  is the Chernoff bound for the event  $\{\prod_j X_j \geq 1\}$  that a probe sees a coordinate spike at unit scale. Convexity of  $\log c_n$  with  $\log c_n(0) = \log c_n(1) = 0$  gives  $0 < \gamma_n$ , and the tangent at 1 gives  $\log c_n(\lambda) \geq -D_n(1 - \lambda)$ , so  $\gamma_n < D_n$ . For  $n = 2$  complex,  $\log c_2(\lambda) = \lambda \log 2 - \log(1 + \lambda)$  is minimized at  $\lambda^* = 1/\log 2 - 1 = 0.4427$ , giving the closed form

$$\gamma_2^{\mathbb{C}} = \log 2 - 1 - \log \log 2 = 0.0596601.$$

Numerically  $\gamma_2^{\mathbb{R}} = 0.109117$ ,  $\gamma_3^{\mathbb{C}} = 0.080129$ ,  $\gamma_4^{\mathbb{C}} = 0.090435$ . Here  $X = \prod_j X_j$  with  $X_j \sim \text{Uniform}[0, 2]$  in the complex binary case.

**Theorem 8.2** (Sharp-exponent bracket). *For  $n \geq 2$  and  $n^q \geq 4$ , for both the real one-column law and the complex block law,*

$$\frac{1}{32}e^{q\gamma_n} \leq \kappa_{\text{opt}}(n, q) \leq e^{qD_n}.$$

Hence  $\gamma_n \leq \liminf_q \frac{1}{q} \log \kappa_{\text{opt}} \leq \limsup_q \frac{1}{q} \log \kappa_{\text{opt}} \leq D_n$ ; for  $n = 2$  complex the optimal per-factor exponent lies in  $[0.0597, 0.1931]$ , and for  $n = 2$  real in  $[0.1091, 0.3069]$ .

*Proof.* The upper bound is Theorem 4.2. We prove the lower bound by constructing a residual with one spike and a flat background.

*Step 1: the residual and its overlap variable.* Set

$$N = n^q, \quad \delta = \frac{1}{N-1}, \quad u = e_1^{\otimes q}, \quad R = \delta I + (1 - \delta)uu^\top.$$

Its eigenvalues are  $1, \delta, \dots, \delta$ , so

$$\operatorname{tr} R = 2, \quad \operatorname{tr} R^2 = 1 + \frac{1}{N-1}, \quad \frac{\operatorname{tr} R^2}{\operatorname{tr} R} \geq \frac{1}{2}.$$

Write

$$X = |u^* z|^2 = \prod_{j=1}^q X_j, \quad X_j = |x_{j,1}|^2.$$

For  $0 \leq \lambda \leq 1$ , independence gives  $\mathbb{E} X^\lambda = c_n(\lambda)^q$ . Since  $\min\{X, 1\} \leq X^\lambda$ ,

$$\mathbb{E} \min\{X, 1\} \leq e^{-q\gamma_n}. \quad (8.1)$$

We will also use

$$\frac{2}{N-1} \leq \frac{4}{N} \leq 4e^{-q\gamma_n},$$

which follows from  $\gamma_n \leq D_n \leq \log n$ .

*Step 2: the real one-column law.* Because  $\|z\|^2 = N$ , the drop is exactly

$$\text{drop} = \frac{(1 - \delta^2)X + \delta^2 N}{(1 - \delta)X + \delta N}.$$

For  $N \geq 4$ ,  $1 - \delta \geq 2/3$  and  $\delta N \geq 1$ . Hence the denominator is at least  $(1 - \delta)(1 + X)$ . Using  $X/(1 + X) \leq \min\{X, 1\}$  and  $\delta^2 N \leq 2/(N - 1)$ , we obtain

$$\mathbb{E} \text{drop} \leq \frac{3}{2} \left( e^{-q\gamma_n} + \frac{2}{N-1} \right) \leq 8e^{-q\gamma_n}.$$

The defining ratio for  $\kappa_{\text{opt}}$  is thus at least  $e^{q\gamma_n}/16$ , which is stronger than required.

*Step 3: the complex two-column block.* Let

$$W = [a \ b], \quad a = \Re z, \quad b = \Im z, \quad G = W^\top W, \quad y = W^\top u.$$

Here  $\|y\|^2 = X$ . Let  $P$  be the range projector of  $R^{1/2}W$ . Since  $R^{1/2}u = u$ , the projector formula gives

$$\begin{aligned} \text{drop} &= \delta \operatorname{rank} P + (1 - \delta) \|Pu\|^2, \\ \|Pu\|^2 &= y^\top (\delta G + (1 - \delta)yy^\top)^\dagger y. \end{aligned}$$

The eigenvalues of  $G$  are

$$\frac{N \pm |z^\top z|}{2}.$$

To bound the smaller one, normalize  $x_j = \sqrt{n} w_j$  and put

$$\Pi = \prod_{j=1}^q |w_j^\top w_j|, \quad |z^\top z| = N\Pi.$$

Independent coordinate phases imply

$$\mathbb{E} |w^\top w|^2 = \sum_{a=1}^n \mathbb{E} |w_a|^4 = \frac{2}{n+1}.$$

Markov's inequality and independence therefore give

$$\mathbb{P}\{\Pi \geq 1/2\} \leq 4 \left( \frac{2}{n+1} \right)^q \leq 4e^{-q\gamma_n}.$$

For the last comparison,  $\gamma_n < D_n < \log((n+1)/2)$ : check  $n = 2$  directly, and use  $D_n^{\mathbb{C}} < 1 - \gamma < \log 2$  for  $n \geq 3$ .

On the event  $\{\Pi < 1/2\}$ ,  $G \succeq (N/4)I$ . Define  $\sigma = y^T G^{-1} y / \delta$ . Sherman–Morrison gives

$$\|Pu\|^2 = \frac{\sigma}{1 + (1 - \delta)\sigma}, \quad \sigma \leq \frac{4X}{\delta N} \leq 4X.$$

It follows that, on this event,

$$\text{drop} \leq 2\delta + \min\{(1 - \delta)\sigma, 1\} \leq 2\delta + 4 \min\{X, 1\}.$$

On its complement, use the deterministic bound  $\text{drop} \leq \text{tr } R = 2$ . Combining these estimates with (8.1),

$$\mathbb{E} \text{drop} \leq \frac{2}{N-1} + 4e^{-q\gamma_n} + 8e^{-q\gamma_n} \leq 16e^{-q\gamma_n}.$$

The numerator in the defining ratio is at least  $1/2$ , so  $\kappa_{\text{opt}} \geq e^{q\gamma_n}/32$ .  $\square$

*Remark 8.3* (What the block buys on the hard family). The proof suggests a bounded-factor advantage for the two-column block: when the real and imaginary parts have squared norms near  $N/2$ , the scalar quotient is approximately  $X/(X + L)$ , whereas the block drop is approximately  $X/(X + L/2)$ , with  $L = \delta(N - 1)$ . Section 10 records a reproducible finite-order check. The proven lower exponent applies to the two-column block; it does not establish an exponent for the larger retained interpolation span.

**Conjecture 8.4.** *For the complex block law at  $n = 2$ ,  $\kappa_{\text{opt}}(2, q) = e^{q\gamma_2 + o(q)}$ ; more generally  $\frac{1}{q} \log \kappa_{\text{opt}}(n, q) \rightarrow \gamma_n$  for both laws.*

If Conjecture 8.4 holds, Theorem 6.4 with  $\kappa = e^{q\gamma_2 + o(q)}$  gives the binary running base  $(2e^{\gamma_2} \cdot 100/81)^{1/3} = 1.378748$  (real branch: 1.401666), an exponential improvement of the upper record. Formally inserting  $\kappa = e^{o(q)}$  into the present sufficient-bound calculation gives  $(200/81)^{1/3} = 1.351600$ . This is a benchmark of an analysis that pays  $N\|R_m\|_F^2/s^2$  with  $\|R_m\|_F^2 \lesssim T^2/m$ ; it is not a lower bound on every algorithm using Nyström and diagonal control. The spike family suggests a possible route: the drift is controlled by the probability that a probe has non-negligible overlap with the dominant direction, for which  $\gamma_n$  gives the Chernoff upper bound used here. Proving a matching rate for every PSD residual requires an additional argument.

## 9 Other probe laws

Lemma 4.1 used only isotropy and a directional moment bound, and Lemma 2.2 only that the drop dominates a quotient. We record three instances. The last two concern ordinary matrix–vector sketches: SRHT rows and CountSketch vectors are not rank-one tensor queries.

## 9.1 Independent sub-Gaussian factors

**Proposition 9.1.** *Let the factors  $x_j \in \mathbb{R}^{n_j}$  be independent and isotropic (coordinates within a factor need not be independent, and discrete laws are allowed), with  $\mathbb{E} \exp(\langle x_j, u \rangle^2 / K_j^2) \leq 2$  for all unit  $u$ . Then  $\mathbb{E} \langle x_j, u \rangle^4 \leq 2K_j^2(K_j^2 - 1)$ , and Theorems 4.2–5.5 and the real branch of Theorem 6.4 hold with*

$$\kappa_{\text{SG}} = \prod_j 2K_j^2(K_j^2 - 1).$$

*Proof.* With  $X = \langle x_j, u \rangle^2$ ,  $e^y \geq 1 + y + y^2/2$  gives  $1 + 1/K^2 + \mathbb{E}X^2/(2K^4) \leq 2$ , i.e.  $\mathbb{E}X^2 \leq 2K^2(K^2 - 1)$ . Tensorize by Lemma 3.1 at  $p = 2$  and apply Lemma 4.1 with  $\beta = 1$ ; Lemma 3.2 is not needed. Zero responses obey the 0/0 convention.  $\square$

Expected PSD progress needs no ordinary Gram inverse, which may fail to exist with positive probability for such laws. For specific laws the near-one limit  $\lim_{\beta} \prod_j m_j(1 + \beta)^{1/\beta}$  is smaller.

## 9.2 A fresh SRHT row

Let  $F$  be an  $n \times n$  orthogonal matrix with flat entries  $|F_{ja}| = n^{-1/2}$  (e.g. a normalized Walsh–Hadamard matrix),  $D = \text{diag}(\epsilon)$  with independent Rademacher signs, and  $B$  any orthogonal matrix independent of  $D$  (deterministic, or itself a random sign-transform layer). A *fresh row probe* is  $z = \sqrt{n} B^\top D F^\top e_J$  for a row index  $J$ , redrawn independently for every probe. This covers the one-layer SRHT ( $B = I$ ) and the two-layer transform  $F D F' D'$ .

**Imported theorem 9.2** (Sharp Khintchine inequality [12]). *For  $r \geq 2$ , real  $a$  and Rademacher  $\epsilon$ ,  $\mathbb{E} |\sum_i a_i \epsilon_i|^r \leq \mathbb{E} |g|^r \|a\|^r$ ,  $g \sim N(0, 1)$ .*

**Proposition 9.3.** *A fresh row probe is isotropic and  $\mathbb{E} |u^\top z|^{2p} \leq g(p) \|u\|^{2p}$  with  $g(p) = 2^p \Gamma(p + 1/2) / \sqrt{\pi}$  for  $p \geq 1$ . Consequently Theorems 4.2–5.5 hold for independent fresh rows with*

$$\kappa_0 = \lim_{\beta \downarrow 0} g(1 + \beta)^{1/\beta} = e^{\log 2 + \psi(3/2)} = \frac{e^{2-\gamma}}{2} = 2.0743278.$$

*This is the dimension-uniform bound obtained by the same moment argument for i.i.d. Rademacher or Gaussian probes in  $\mathbb{R}^N$ , and the  $n \rightarrow \infty$  limit of  $e^{D_n^{\mathbb{R}}}$ . It need not be the best constant at a fixed finite dimension.*

*Proof.*  $u^\top z = \sum_a \epsilon_a (\sqrt{n} F_{Ja}) (Bu)_a$  is a Rademacher sum with coefficient norm  $\|Bu\| = \|u\|$ ; apply Imported Theorem 9.2 with  $r = 2p$  conditionally on  $B$ . Isotropy:  $\mathbb{E} z z^\top = n B^\top \text{diag}(F_j^2) B = I$ . Since  $g(1) = 1$ ,  $\kappa_0 = \exp\{(\log g)'(1)\} = \exp\{\log 2 + \psi(3/2)\}$  and  $\psi(3/2) = 2 - \gamma - 2 \log 2$ . For i.i.d. Rademacher probes the directional moments are bounded by the same  $g(p)$ ; flat directions approach Gaussian moments as the dimension grows. Thus the dimension-uniform bounds coincide; and  $D_n^{\mathbb{R}} = \log n + \psi(3/2) - \psi(n/2 + 1) \rightarrow \log 2 + \psi(3/2)$ .  $\square$

Thus the same dimension-uniform sufficient drift constant applies to a fresh SRHT row and to an unstructured Rademacher probe. This comparison of upper bounds does not establish equality of their optimal drift constants. The rows of one SRHT sketch share  $D$ ; after one row is used the residual is correlated with the others, so the  $M$  rows of a single sketch are *not*  $M$  independent updates, and this proposition does not count them as such.

### 9.3 Whole CountSketch blocks

Let  $H \in \mathbb{R}^{b \times N}$  be a CountSketch block: independent uniform hashes  $h_i \in [b]$  and independent signs  $\sigma_i$ ,  $H_{h_i, i} = \sigma_i$ . The update is  $R' = R - RH^\top(HRH^\top)^\dagger HR$ .

**Lemma 9.4** (Mixed trace identity). *For symmetric  $B, R$  and  $X_B = \text{tr}(HBH^\top)$ ,  $\mathbb{E}X_B = \text{tr } B$  and*

$$\begin{aligned} \mathbb{E}(X_B X_R) &= \text{tr } B \text{tr } R + \frac{2}{b} \left( \text{tr}(BR) - \sum_i B_{ii} R_{ii} \right) \\ &\leq \left( 1 + \frac{2}{b} \right) \text{tr } B \text{tr } R \quad (B, R \succeq 0). \end{aligned}$$

*Proof.*  $X_B = \text{tr } B + \sum_{i \neq j} B_{ij} \sigma_i \sigma_j \mathbf{1}_{h_i = h_j}$ . In the product of two such sums only matching unordered pairs survive the sign expectation, each with hash probability  $1/b$ , giving  $(2/b) \sum_{i \neq j} B_{ij} R_{ij}$ . For PSD matrices  $\text{tr}(BR) \leq \text{tr } B \text{tr } R$  and  $B_{ii} R_{ii} \geq 0$ .  $\square$

**Theorem 9.5** (CountSketch block drift). *With  $\kappa_b = 1 + 2/b$ , for every nonzero PSD  $R$  and every finite convex  $f \geq 0$  with  $f(0) = 0$ ,*

$$\mathbb{E}(\text{tr } R - \text{tr } R') \geq \frac{\text{tr } R^2}{\kappa_b \text{tr } R}, \quad \mathbb{E} \text{tr } f(R') \leq \text{tr } f(R) - \frac{\text{tr}[Rf(R)]}{\kappa_b \text{tr } R}.$$

Hence Theorems 5.2–5.5 hold for  $s$  independent blocks with  $\kappa_b$  in place of  $\kappa$  and  $s$  counting blocks.

*Proof.* Let  $F = R^{1/2} H^\top H R^{1/2}$  and  $P$  its range projector. Each eigenvalue of  $F$  is at most  $\text{tr } F = X_R$ , so  $F \preceq X_R P$  and  $\text{tr}(PR) \geq \text{tr}(FR)/X_R = X_R^2/X_R$ , and likewise  $\text{tr}(Pf(R)) \geq X_{Rf(R)}/X_R$ . Under  $dQ = X_B dP/\text{tr } B$ , Jensen gives  $\mathbb{E}[X_B/X_R] \geq (\text{tr } B)^2/\mathbb{E}(X_B X_R) \geq \text{tr } B/(\kappa_b \text{tr } R)$  (Lemma 9.4;  $X_R = 0$  forces  $X_B = 0$  when  $\ker R \subseteq \ker B$ ). The compression argument of Theorem 5.3 finishes the proof.  $\square$

*Remark 9.6* (Scope: one block is credited like one probe). The bound uses  $F \preceq (\text{tr } F)P$  and so credits a  $b$ -column block with the progress of a single normalized probe. For rank-one residuals it is within  $\kappa_b$  of the maximal possible drop. Already for  $R = I_N$  with  $N \gg b$ , it loses a factor of order  $b$ : the true drop is the number of nonempty buckets, about  $b$ , while the bound gives  $1/\kappa_b$ . In sketch-column count these guarantees are therefore weaker than per-column bounds by up to a factor  $b$ ; their merits are the explicit constant and the use of independence only across blocks.

## 10 Reproducible numerical checks

The accompanying [verification script](#) and [machine-readable results](#) recompute the constants and budgets in double precision. They also run a seeded Monte Carlo experiment and check selected matrix identities. These checks support the calculations; they do not replace the proofs.

### 10.1 Constants and integer budgets

The two binary order-one constants are

$$e^{D_2^{\mathbb{R}}} = 1.3591409142, \quad e^{D_2^{\mathbb{C}}} = 1.2130613194.$$

A central finite difference for  $(\log c_n)'(1)$ , for both fields and  $n \in \{2, 3, 5, 10\}$ , agrees with  $D_n$  within  $1.1 \cdot 10^{-10}$ . Numerical minimization over  $[0, 1]$  gives

$$\gamma_2^{\mathbb{R}} = 0.1091170039, \quad \gamma_2^{\mathbb{C}} = 0.0596601011.$$

The latter agrees with the closed form in Section 8. The leading comparison ratios are 18.441745 and 44.230199.

Table 1 uses  $n = 2$ ,  $\varepsilon = (9/10)^q$ . All counts are uncapped; the algorithm may instead use  $N = 2^q$  calls. The script evaluates the displayed integer formulas. For large ranges of the auxiliary rank, it searches around the continuous minimizer; the table is a numerical optimization check, not an independent certificate of the global integer optimum.

| $q$ | real drift | complex drift | direct diagonal control |
|-----|------------|---------------|-------------------------|
| 8   | 138        | 214           | 184                     |
| 12  | 693        | 1,039         | 1,112                   |
| 16  | 3,474      | 4,880         | 6,770                   |
| 20  | 17,457     | 22,583        | 41,264                  |
| 24  | 87,699     | 103,261       | 251,564                 |
| 30  | 987,645    | 995,187       | 3,786,870               |
| 40  | 55,885,335 | 42,312,585    | 347,540,288             |

| $q$ | fourth-moment benchmark | fractional benchmark | empirical fractional |
|-----|-------------------------|----------------------|----------------------|
| 8   | 37,984                  | 1,789,197            | 2,916                |
| 12  | 193,348                 | 8,169,539            | 30,875               |
| 16  | 985,320                 | 33,776,624           | 296,395              |
| 20  | 5,027,762               | 131,952,535          | 2,688,105            |
| 24  | 25,694,427              | 497,015,252          | 23,495,275           |
| 30  | 271,702,808             | 3,474,278,491        | 579,594,817          |
| 40  | 7,844,091,534           | 83,059,865,691       | 111,965,394,931      |

Table 1: Explicit uncapped query budgets. The two panels separate the new drift construction from the comparison formulas.

The exact drift integers switch from the real branch to the complex branch at  $q = 31$ , both for  $\varepsilon = (9/10)^q$  and for  $\varepsilon = 0.1$ , in the checked range  $2 \leq q \leq 60$ . This agrees with the leading-term crossover of Proposition 6.5.

## 10.2 The spike-plus-flat family

We draw 20,000 independent complex-spherical product probes for each tensor order, using the random seed 20260929. The input is the family in the proof of Theorem 8.2, with  $n = 2$ . Table 2 reports the expected drop divided by  $\text{tr } R^2 / \text{tr } R$ ; parentheses give one estimated standard error. The experiment uses exact two-by-two formulas for the block Gram matrix and does not expand the ambient tensor.

The observed block/quotient ratios range from approximately 1.33 to 1.51. Each observed progress value exceeds the proved sufficient bound  $1/\kappa_{\mathbb{C}}$ . This finite-order experiment neither

| $q$ | scalar quotient | two-column block | $1/\kappa_{\mathbb{C}}$ | $e^{-q\gamma_2}$ |
|-----|-----------------|------------------|-------------------------|------------------|
| 4   | 0.713 (0.003)   | 1.077 (0.004)    | 0.462                   | 0.788            |
| 6   | 0.581 (0.004)   | 0.808 (0.004)    | 0.314                   | 0.699            |
| 8   | 0.483 (0.004)   | 0.650 (0.004)    | 0.213                   | 0.620            |
| 10  | 0.404 (0.004)   | 0.541 (0.004)    | 0.145                   | 0.551            |
| 12  | 0.337 (0.004)   | 0.451 (0.004)    | 0.098                   | 0.489            |
| 14  | 0.283 (0.003)   | 0.380 (0.004)    | 0.067                   | 0.434            |

Table 2: Seeded spike-family experiment. The Chernoff column is a rate benchmark, not a claimed lower bound on the expected drop.

establishes the conjectured asymptotic exponent nor evaluates the larger retained interpolation span.

### 10.3 Algebraic spot checks

For PSD matrices of dimensions 4, 8, 12, including singular inputs, the sequential-update identity in Lemma 2.1 agrees with a direct pseudoinverse calculation to Frobenius error below  $9 \cdot 10^{-15}$ . The numerical construction includes rank-one, intermediate-rank, and full-rank matrices. These small checks are diagnostics of the implementation, not a proof of the identity.

## 11 Open problems

1. **Close the exponent bracket.** Prove Conjecture 8.4: a drift constant  $e^{q\gamma_n + o(q)}$  for the complex block (or the real probe). At  $n = 2$ ,  $\varepsilon = (9/10)^q$  this lowers the upper base from 1.441482 to 1.378748. A natural route is a mixed-moment argument that keeps the event  $\{|u^*z|^2 \gtrsim 1\}$  rather than the average log-moment.
2. **Improve the analysis benchmark.** The present variance and Frobenius bounds give base 1.351600 if one formally removes the exponential drift penalty. This is a sufficient-bound benchmark, not an algorithmic impossibility result. Better residual-specific diagonal bounds, or adaptive pivoting after a probe, could improve it.
3. **Block-level mixed moments.** Replace  $\text{tr } F$  by  $\lambda_{\max}(F)$  in the CountSketch and complex-block arguments to credit a block with more than one probe’s progress.
4. **Retention.** Quantify the gain of the retained  $(q+1)$ -span; the numerics suggest a constant factor only, which should be proved.
5. **Stability and bits.** Numerically stable interpolation nodes and a polynomial-size discretization of the angular laws (Appendix A is existential).
6. **Relative spectral error.** A rank- $r$  relative operator-norm analogue of Theorem 5.5 for product probes, without OSE-level sketch sizes.

## A Finite randomness with arbitrarily small slack

**Lemma A.1** (Finite isotropic angular laws). *Fix  $n, p > 1$  and  $\delta > 0$ . There is a finite law of nonzero rational raw vectors over  $\mathbb{R}$ , or Gaussian-rational raw vectors over  $\mathbb{C}$ , with dyadic probabilities, whose formal normalization  $x = \sqrt{n}g/\|g\|$  satisfies  $\mathbb{E}xx^* = I$  and  $\sup_{\|u\|=1} \mathbb{E}|u^*x|^{2p} \leq c_n(p) + \delta$ .*

*Proof.* Approximate a standard Gaussian scalar by symmetric finite laws on nonzero rational half-grid points with dyadic probabilities, converging weakly. Use i.i.d. coordinates (two copies per coordinate in the complex case). Coordinate permutations and sign changes (and multiplication by  $i$  in the complex case) are symmetries, so the normalized covariance is scalar with trace  $n$ : exact isotropy. Normalization is continuous away from 0, where the Gaussian has no atom, so the angular laws converge weakly to the uniform sphere law. The functions  $x \mapsto |u^*x|^{2p}$ ,  $\|u\| = 1$ , are bounded and equicontinuous on the compact sphere; a finite net in  $u$  gives uniform convergence of their expectations.  $\square$

This follows the angular construction of [4, Section 6].

**Proposition A.2** (Finite-law drift). *For every  $\eta > 0$  the finite laws can be chosen so that all drift-based bounds hold with  $(1 + \eta)\kappa$ .*

*Proof.* Choose  $\beta > 0$  with  $H(1 + \beta)^{1/\beta} < (1 + \eta)\kappa$  and then local tolerances  $\delta_j$  with  $[\prod_j (c_{n_j}(1 + \beta) + \delta_j)]^{1/\beta} \leq (1 + \eta)\kappa$ . Tensorization and isotropy hold for the finite laws, and Lemma 4.1 at this  $\beta$  gives drift with  $(1 + \eta)\kappa$ . At a zero denominator the numerator vanishes. The algorithm queries raw rational factors; normalization rescales a real column or a whole complex block by one positive scalar, preserving the Nyström output. Interpolation uses rational nodes; the other batches use fair signs and the bounded-bit coordinate law.  $\square$

This preserves the quantitative guarantees, not almost-sure rank recovery.

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